

RENEE CRISPO

PhD Student in Applied Mathematics at INRIA, Paris

renee.crispo@inria.fr

EDUCATION

- Jan 2026 – present **PhD in Applied Mathematics**
COMMEDIA team, Inria, Paris, France
Thesis title: Modeling and Simulation of 3D Cardiac Hemodynamics in Hypoplastic Left Heart Syndrome
Supervisor: Miguel A. Fernández
- Sep 2023 – Dec 2025 **Master of Science degree in Mathematical Engineering**
Politecnico di Milano, Milano, Italy
Thesis title: Numerical analysis of loosely coupled systems for Fluid-Structure Interaction under large added-mass effects.
Supervisors: Christian Vergara and Miguel A. Fernández
Final mark: 110/110
- Sep 2024 – Dec 2025 **Master of Science degree in Mathématiques de la Modélisation**
Sorbonne Université, Paris, France
Double degree program between PoliMi and Sorbonne.
- Sep 2020 – Sep 2023 **Bachelor of Science degree in Mathematical Engineering**
Politecnico di Milano, Milano, Italy
Main courses: Calculus, PDEs, Numerical Analysis, Probability, Statistics, Mechanics, Physics, Computer Science.
Final mark: 108/110
- Sep 2018 – Jun 2019 **Exchange student during fourth year of high school**
Saint Mary's Bay Academy, Weymouth, Nova Scotia, Canada
Awarded as the best 12th grade student in pre-calculus, calculus, biology, and PE leadership.
- Sep 2015 – Jun 2020 **High school diploma**
Liceo Scientifico Leonardo da Vinci, Gallarate, Italy
Final mark: 98/100

EXPERIENCE

- Apr 2025 – Jul 2025 **Research Intern**
Four-month internship in COMMEDIA team at Inria, Paris, France
Title: On the numerical Analysis of loosely-coupled schemes for Fluid-Structure Interaction
Supervisor: Miguel A. Fernández
- Sep 2021–Jul 2024 **Tutoring**
Private lessons of math, physics and calculus for middle and high school students.

TALKS AND PRESENTATIONS

- Jun 2026 **COMMEDIA, Inria Seminar**

PROJECTS

UNIVERSITY PROJECTS:

- Mar 2025 **Numerical Analysis of blood flow in a bifurcation and of explicit schemes for FSI**
Project in FreeFem++ for the course of *Modèles mathématiques et méthodes numériques pour la simulation en hémodynamique* at Sorbonne. [Repo](#).
- Jan 2025 **Implementation of a Navier-Stokes solver with boundary conditions in C and FreeFem++**
Project in C and Python for the course of *From EDP to their resolution by Finite Elements* at Sorbonne. [Repo](#).
- Dec 2024 **Domain Decomposition Methods: analysis and implementation of the Helmotz equation**
Project in Python and MPI for the course of *HPC for Numerical Methods and Data Analysis* at Sorbonne. [Repo](#).
- Nov 2024 **A comparative analysis of MGS and TSQR algorithms in parallel environment**
Project in Python and MPI for the course of *HPC for Numerical Methods and Data Analysis* at Sorbonne.
- Apr 2024 – Sep 2024 **Isogeometric analysis of high-order PDEs**
Discretisation and analysis of the Cahn-Hilliard equation focusing on its application in tumor growth, using the GeoPDEs library in MatLab for the course of *Numerical Analysis for PDEs* at PoliMi. [Repo](#).

AWARDS AND SCHOLARSHIPS

- 2022 – 2025 **High-merit scholarship**
Politecnico di Milano: scholarship for students with particularly high merit (mark average over 27/30)
- 2022 **Best Freshman**
Politecnico di Milano: award based on the academic achievement of the first year of the Bachelor degree

LANGUAGES

Italian: native. **English:** C1. **French:** B1.

TECHNICAL STRENGTHS

Programming languages: MatLab, C++, C, Python, R, Mathematica.

Scientific writing: LaTeX, Beamer.

Scientific tools: NumPy, Matplotlib, MPI, FreeFem++.